MU CHUPING

Technical AI Safety and Machine Learning

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Cambridge, UK

Interests: evaluation and oversight of autonomous agents; reproducible ML pipelines.

EDUCATION

University of Cambridge (Trinity Hall) — BA Natural Sciences | October 2026 – 2029 (expected)
Part IA: Mathematics (Course B), Chemistry, Biomedical Sciences, Earth Sciences
National Science Scholarship, A*STAR

NUS High School of Math and Science — CAP 4.74/5.0, High Distinction | Graduated 2023

EXPERIENCE

Genome Institute of Singapore, A*STAR — Wan Lab | Student Intern | 9 February – 4 September 2026

- Rewrote cellSNP (single-cell allele pileup) in C++17/HTSlib with parallel contig partitioning. On a 3.2 GB BAM in mode 2 it ran in 7.8 minutes on one process against 7.2 hours on ten, using 7.6x less memory; gains vary by configuration.
- Built a fail-closed equivalence harness before trusting the speedup: 63,180 parameter cases plus a full 62.4M-record BAM, each requiring exact matches across six output files, with hash-checked repeat runs.
- Benchmarked PCA against PyTorch VAEs over five seeds on 80-cell RNA-structure profiles. Label-free PCA + Gaussian mixture separated both cell lines (ARI 0.95), beating the best VAE (silhouette 0.56 vs 0.18). Also found label leakage in earlier probe-based metrics.
- Tested whether single-cell RNA structure-probing analysis transfers to spatial transcriptomics, building a gated, resumable pipeline (Window) and running it end to end on mouse-brain data.

Trend-Pact | Co-founder and COO | April 2025 – present

- Built the software that runs booking, check-in, door access, cleaning and guest support for 94 rooms across two Osaka properties, now on a paying monthly contract.
- Added an LLM guest-support assistant with structured-output checks that escalates to staff when it is unsure.
- Evaluated passport-selfie face verification against a preregistered criterion: fusing SFace and AuraFace scores reached 98.4% genuine acceptance at 0.09% false-match rate, but cut false matches only from 245 to 198 against SFace alone, which was not statistically significant.

PROJECTS

Biohub Cell Tracking During Development | Kaggle, solo | July – August 2026

- Reproduced a public 3D U-Net + node-transformer baseline, then screened fine-tuning, PEFT and graph/ILP association variants under grouped cross-validation with preset promotion gates. None beat the baseline; private score 0.898 (public 0.915).
- Ran the campaign through Codex and Claude Code. Review steps caught agent errors before they used GPU time, including a release plan naming the wrong model and a memory estimate that omitted a 5.2 GB model footprint.

cruise | Open-source plugin | July 2026

- A plugin for Codex and Claude Code that makes an agent produce fresh evidence before it can mark a task done. I built it because my own agents kept reporting unfinished work as complete.

OTHER

- **National Service**, Singapore Armed Forces (2024 – 2026): armoured infantryman; helped run camp logistics with regular staff in my second year.
- **Early research**: two Springer conference-proceedings chapters (2023), one as first author, on COVID-19 modelling and allergen profiling. Invited by the Singapore Mathematical Society to present the SEIRQV model and its use in teaching.
- **Awards**: IChTo Silver and Best Solution (2023); Singapore Chemistry Olympiad Gold, Singapore Math Project Festival Gold and SSEF Yale-NUS Award (2022).
- **Skills**: Python, C++17 (HTSlib), TypeScript, SQL, R; PyTorch, NumPy/pandas, scikit-learn; Slurm, Git; LLM APIs and coding agents.